New Brunswick, Canada · 2026

# New Brunswick Responsible AI Framework

A practitioner-built, community-owned governancestandardforsecure,trustworthy, and accountable AI — built for NB's people, not just its technology.

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|---|---|
| *VERSION* | Draft v1.1 — Open Consultation |
| *STATUS* | Voluntary Standard — Not Enacted Legislation |
| *LICENCE* | Creative Commons Attribution 4.0 International (CC BY 4.0) |
| *ALIGNED TO* | EU AI Act · NIST AI RMF 1.0 · ISO 42001 · Bill C-8 · OCAP® · UNDRIP · SANS AISMM (May 2026) |
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***VERSION STATUS — OPEN FOR PUBLIC CONSULTATION***

This is a practitioner-drafted voluntary standard, not enacted provincial legislation. It operates as a procurement condition, professional baseline, and policy proposal pending provincial legislative action and formal consultation. It will be revised following structured consultation with First Nations communities, labour, civil society, and the NB technology sector. Where AI systems engage federal jurisdiction, this framework complements, and does not supplant, federal obligations.

***RELATIONSHIP TO THE EU AI ACT — ADAPTATION, NOT ADOPTION***

This framework draws on the EU AI Act's risk-tiered architecture as a starting point — not a wholesale import. The EU AI Act was designed for a single market with pan-European enforcement bodies that Canada does not yet have. Where the EU model is adapted here, it is calibrated to NB's provincial jurisdiction and the operational realities of NB's bilingual, SME-driven, public-service-reliant economy.

*§ 1 · PREAMBLE*

# Why New Brunswick Needs Its Own AI Framework

Artificial intelligence is being deployed acrossNewBrunswick'spublicand privatesectors atan accelerating pace — from government service chatbots and cybersecurity threat detection to sentiment analysis tools used by healthcare and social service providers. Without a clear, jurisdiction-specific governance framework, these deployments risk causing harm to citizens, eroding public trust, and exposing organizations to legal and reputational liability.

This framework provides New Brunswick consultants, technology vendors, public sector bodies, and regulated organizations a practical, principled roadmap for deploying AI systems that are **secure, transparent, trustworthy, fair, reliable, traceable, explainable, and accountable** — anchored in the lived realities of NB's bilingual, geographically diverse, and public-service-reliant population.

## SCOPE OF APPLICATION

- Provincial and municipal government services and digital transformation projects
- Cybersecurity operations including AI-driven threat detection and incident response
- AI chatbots and virtual assistants interfacing with NB residents
- Sentiment analysis tools applied to social media, call centre, or public feedback data
- Healthcare, social services, and education AI applications
- IT consultancies deploying AI under CyberSecure NB or related provincial programs

*BILINGUALISM REQUIREMENT*

All AI systems deployed in public-facing NB contexts must respect the Official Languages Act. AI outputs, explanations, and human oversight mechanisms must be available in both English and French. Sentiment analysis tools must be validated against both language corpora to avoid disparate accuracy across linguistic communities.

## FIRST NATIONS PEOPLES: RIGHTS-HOLDERS, NOT JUST PROTECTED CLASSES

New Brunswick is home to fifteen First Nations communities across two nations — Wolastoqiyik (Maliseet) and Mi'kmaq peoples — whose rights predate and exist independently of provincial governance structures. This framework explicitly recognizes that First Nations communities in NB are **rights-holders and governance participants** in AI deployment decisions affecting their peoples, territories, or data — not merely a demographic category to be protected from algorithmic bias.

*OCAP® PRINCIPLES — MANDATORY FOR AI SYSTEMS INVOLVING FIRST NATIONS DATA*

Any AI system that collects, processes, stores, or generates inferences from data about NB First Nations communities must be governed by the OCAP® principles: Ownership (the community owns its data collectively) · Control (the community controls how data is used) · Access (the community has access to data held about it) · Possession (the community maintains physical custody of data). These are not aspirational — they are the established Indigenous data sovereignty standard in Canadian law and practice.

— **Nation-to-Nation consultation required:** AI systems affecting First Nations communities must be developed in consultation with the relevant Nation(s) prior to procurement — not as a post-deployment notification

— **Free, Prior and Informed Consent (FPIC):** Where AI systems could affect First Nations rights, lands, or governance practices, FPIC consistent with UNDRIP applies

— **No extractive AI:** AI systems must not extract, aggregate, or commercialize knowledge, language, cultural expressions, or community data without explicit consent from the relevant Nation's governance body

— **Indigenous language AI:** Wolastoqey and Mi'kmaw languages are living languages. AI systems that process Indigenous language content must be developed with, not merely for, the relevant language communities

— **Framework co-development commitment:** This framework will be revised following structured Nation-to-Nation consultation. First Nations input will materially shape subsequent versions.

***ACKNOWLEDGEMENT OF LIMITATIONS***

This v1.1 draft was developed by a GRC practitioner, not in co-design with First Nations communities. That is a limitation that must be named, not obscured. The framework will not be complete until it has been shaped by Wolastoqiyik and Mi'kmaq governance bodies directly. We actively invite that engagement.

### BILINGUAL AI — INTERNATIONAL REFERENCE: INDIA'S SUVAS MODEL

Canada's bilingual governance challenge in NB has a useful international parallel in India's SUVAS (Supreme Court Vidhik Anuvaad Software) — an AI-powered machine-assisted translation system launched in 2019 by India's Supreme Court to translate judicial documents between English and eleven Indian vernacular languages.

**SUVAS is not directly applicable** to NB's English-French bilingual requirement — it was designed for a different language family and legal domain. However, its **governance architecture** is precisely what NB should require for any AI translation used in legal or government contexts:

— **Human-in-the-loop post-editing:** SUVAS generates AI draft translations reviewed and refined by trained human experts (including retired judges and senior law clerks) before publication — no raw AI translation is used for consequential legal text

— **Domain-specific training corpus:** Models are trained exclusively on judicial documents — judgments, orders, statutes, pleadings — not general-purpose text, ensuring domain accuracy

— **Continuous feedback loop:** Corrections made during human post-editing are fed back into the system for incremental accuracy improvement over time

— **Scale and validation:** By 2023, SUVAS had translated over 31,000 of the Supreme Court's 36,000 judgments into scheduled languages, demonstrating the viability of AI-assisted bilingual governance at scale

*NB BILINGUAL AI GOVERNANCE — SUVAS LESSON APPLIED*

The SUVAS model directly informs NB's bilingual AI requirement: any AI system generating text, decisions, or explanations in English or French for NB residents must follow the SUVAS governance pattern — AI drafts, human domain experts validate, errors improve the model, and no raw AI output is used for consequential administrative or legal text without human review. This applies equally to Acadian French dialect content, which requires validation by Acadian French language experts, not just standard French speakers.

### CYBERSECURE NB: THE PROVINCIAL AI & CYBERSECURITY GRANT PROGRAM

The **CyberSecure NB** program, delivered through the **CBDC** network, reimburses eligible SMEs up to **$3,000** for projects integrating cybersecurity and AI into business operations — the primary vehicle through which this framework's principles reach NB's SME ecosystem.

| *DETAIL* | *SPECIFICS* |
|---|---|
| Delivery | Administered by local CBDC offices across New Brunswick |
| Reimbursement | Up to $3,000 for eligible professional services and implementation |
| Eligibility | For-profit entity registered in NB with at least $30,000 annual revenue |
| How it works | Engage a service provider · pay · submit invoice + report to CBDC |
| Commitment | Maintain cybersecurity or AI strategy for at least six months after participation |

*DEADLINE NOTICE*

The application deadline for some streams was previously March 31, 2026. Contact your local CBDC office directly to confirm current availability.

### SME RECIPIENT OBLIGATIONS UNDER CYBERSECURE NB

A critical governance gap: the CyberSecure NB program scrutinizes eligible costs but not responsible deployment. When an NB SME deploys an AI system, that business becomes the **operator**. The consultant may have left; the vendor may be offshore. The SME owner makes consequential decisions with AI outputs — and its customers, employees, or community members bear the consequences if it fails.

#### SME Baseline Obligations

01 **Know your risk tier.** Confirm in writing which risk tier your AI system falls under before your consultant leaves.

02 **Understand what data your AI processes.** What data does the system collect? Is it stored in Canada? Who has access?

03 **Maintain a human decision-maker.** For any AI output affecting a customer or employee — the AI advises; you decide.

04 **Keep the vendor honest.** Your AI vendor's contract must include: what the system does, what happens when it fails, and how you exit if it causes harm.

05 **Tell people.** If your business uses AI to screen applications, respond to inquiries, or analyze feedback — disclose it.

| AI USE CASE | RISK TIER | SME MINIMUM OBLIGATIONS |
|---|---|---|
| AI Chatbot (customer service) | Tier 2 | Disclose AI use · offer human escalation · log interactions · monthly output review |
| Sentiment analysis (feedback) | Tier 2 | Inform subjects · do not use as sole basis for decisions · retain audit log |
| AI-assisted hiring / recruitment | Tier 1 | Full human review · bias audit · candidates informed AI was used |
| Predictive inventory / forecasting | Tier 3 | Document the system · validate outputs periodically against actuals |
| AI bookkeeping / accounting | Tier 3 | Human sign-off on financial outputs · data must remain in Canadian jurisdiction |

### SCOPE BOUNDARIES: WHAT THIS VERSION DOES NOT COVER

| OUT-OF-SCOPE AREA | WHY IT MATTERS | PLANNED |
|---|---|---|
| Environmental harms | AI energy consumption implicates NB Power's grid and NB's climate commitments | v2.0 — NB Power input required |
| Labour displacement | Workers displaced by AI-funded automation deserve retraining and notification obligations | v2.0 — NBFL and PSAC engagement |
| Youth & children's AI safety | Relational dependence, cognitive impact, harmful content in youth-facing AI tools | v2.0 — Education NB input |
| AI and democratic integrity | Generative AI disinformation and electoral interference in NB political environment | v2.0 — Elections NB engagement |

*§ 2 · FOUNDATIONAL PRINCIPLES*

# Nine Core Principles of NB Responsible AI

Theseprinciplesarethe testablestandards againstwhichAI systemsdeployed in NB must be assessed. Two principles — **Indigenous Sovereignty** and **Environmentally Conscious** — were added in v1.1 following initial scrutiny review.

| PRINCIPLE | STANDARD | VER |
|---|---|---|
| Secure by Design | AI systems must be architected with cybersecurity controls embedded from inception, aligned with CyberSecure NB zero-trust principles and NIST CSF 2.0 | Core |
| Transparent | Users must be informed when interacting with an AI system. Operators must document training data, decision influence, and output generation | Core |
| Fair & Non-Discriminatory | AI systems must be tested for bias across NB Human Rights Act protected grounds, including language, race, gender, disability, and age — across both official language communities | Core |
| Reliable & Robust | Systems must perform consistently across edge cases, adversarial inputs, and operational changes. Failure modes must be identified before deployment | Core |
| Traceable | All AI-influenced decisions must be auditable. Logs of model inputs, outputs, versions, and human overrides must be retained and accessible to oversight bodies | Core |
| Explainable | Affected individuals and oversight bodies must receive plain-language explanations of AI outputs. XAI requirements are proportional to risk tier | Core |
| Accountable | A named individual or body must bear legal and operational accountability for each AI system. Responsibility cannot be delegated to a vendor or the AI system itself | Core |
| Indigenous Sovereignty | First Nations peoples in NB are rights-holders in AI governance. AI systems affecting Wolastoqiyik and Mi'kmaq communities must respect OCAP® principles, FPIC, and UNDRIP — and be developed with, not merely for, those communities | v1.1 |
| Environmentally Conscious | AI deployment decisions must acknowledge energy consumption and carbon implications. Large-scale AI infrastructure choices carry environmental costs that are a legitimate subject of governance. Addressed fully in v2.0 | v1.1 |

*§ 3 · RISK CLASSIFICATION*

# AI Risk Tier Model for New Brunswick

Adaptedfrom theEU AIAct's four-tier riskhierarchy,tailored for NB's public sector, consultancy, and cybersecurity operational contexts. Risk tier determines conformity requirements, documentation obligations, and human oversight thresholds.

| TIER | LEVEL | NB EXAMPLES | KEY OBLIGATIONS |
|---|---|---|---|
| Tier 0 — PROHIBITED | Banned Outright | Real-time biometric mass surveillance of NB citizens; social scoring systems; subliminal manipulation | Banned outright. No deployment permitted. Cannot be procured by NB public bodies. |
| Tier 1 — HIGH RISK | Mandatory Full Oversight | AI in child welfare; parole/sentencing tools; CI protection AI; automated government hiring; healthcare diagnostics | Mandatory conformity assessment, bias audit, human-in-the-loop, registration, incident reporting, DPIA required |
| Tier 2 — LIMITED RISK | Transparency & Logging | GNB portal chatbots; citizen sentiment analysis; AI-assisted cybersecurity triage; procurement advisory AI | Transparency obligations required; logging and audit trail; human review of flagged outputs; annual review cycle |
| Tier 3 — MINIMAL RISK | Voluntary Standards | AI spam filters; grammar/translation tools; internal knowledge management; code review AI for internal teams | Voluntary code of conduct. Basic documentation recommended. No mandatory conformity assessment. |

**PRACTITIONER NOTE — WHEN IN DOUBT, CLASSIFY UP**

Correct risk tier classification is one of the most consequential decisions in any AI engagement. Deploying a Tier 1 system under Tier 2 obligations — whether through misunderstanding or client pressure — exposes NB residents to unmitigated harm and the practitioner to serious professional and reputational consequences.

*§ 4 · IMPLEMENTATION PILLARS*

# Five Pillars of NB AI Implementation

Thesepillars translatethe nineprinciples into actionabletechnicaland governance requirements. Each pillar has a designated accountability owner within the deploying organization.

**1**

SECURITY ARCHITECTURE

### Cybersecurity-First AI Design

Accountability: CISOor designatedAISecurity Lead

→ Conduct AI-specific threat modelling (prompt injection, model inversion, data poisoning, adversarial inputs) before deployment

→ Apply zero-trust access controls to AI model APIs, training pipelines, and inference endpoints

→ Implement runtime monitoring for model drift, anomalous output patterns, and unauthorized access attempts

→ Maintain a **Model Card** and **System Card** for every deployed model, updated at each version change

→ For RAG architectures and AI chatbots: validate retrieval sources, apply output filtering, and log all retrieval-to-response chains

→ Perform penetration testing of AI interfaces at minimum annually (quarterly for Tier 1 systems)

→ **Data residency:** AI systems processing NB public sector, health, or First Nations data must default to Canadian data residency

→ **Foundation model governance:** Deployers using commercial foundation models remain fully responsible for outputs — 'we're just calling an API' does not transfer accountability to the model vendor

**2**

TRANSPARENCY & DISCLOSURE

### Honest AI Interaction Standards

Accountability: CommunicationsorDigitalServices Lead

→ All chatbot and virtual assistant interfaces must display a visible, plain-language disclosure in both English and French

→ Automated decisions that materially affect a person must be identified as AI-generated, with a human review pathway offered

→ Sentiment analysis tools must disclose to data subjects that their communications are subject to automated scoring, where legally required

→ System documentation must be available to oversight bodies upon request within 30 days

→ AI-generated content used in government communications must be disclosed as such

3

FAIRNESS & BIAS MANAGEMENT

## Equitable AI for All New Brunswickers

Accountability: EquityOfficer, legal counsel,ordesignated bias auditor

→ Conduct pre-deployment bias assessments across all NB Human Rights Act protected grounds including race, national origin, age, disability, sex, and gender identity

→ Validate model performance parity across English and French language inputs — especially for sentiment analysis and NLP models

→ Test for disparate impact on First Nations communities, rural residents, and seniors

→ Maintain a Bias Incident Register; review quarterly; remediate within defined SLAs

→ Do not use postal code, language, or community affiliation as proxy variables for protected characteristics

4

EXPLAINABILITY & AUDITABILITY

## Traceable AI Decision Chains

Accountability: DataGovernance Officeror AI Compliance Lead. Log retention: 5 years (Tier 1), 2 years (Tier 2)

→ Implement structured audit logs: input → model version → output → any human override, with timestamps and operator identifiers

→ For high-risk decisions: provide human-readable explanation (SHAP, LIME, or equivalent XAI) to affected parties within 10 business days of request

→ Maintain version control of all deployed models; document what changed between versions and why

→ Conduct post-deployment monitoring: track accuracy, drift, and outcome disparity on a rolling basis

→ Establish an AI Incident Reporting process: define what constitutes an AI incident, escalation path, and public disclosure thresholds

5

GOVERNANCE & HUMAN OVERSIGHT

## Keeping Humans Accountable

Accountability: Named AI Accountability Owner — designated at Deputy Minister or C-suite level. Cannot be delegated to a vendor, technical team, or the AI system itself

→ Designate an **AI Accountability Owner** (named individual) for every deployed system — accountable for compliance, incidents, and remediation

→ Establish an AI Review Board with representation from legal, privacy, equity, and technical functions

→ Require human-in-the-loop review for all Tier 1 AI outputs before action is taken on a citizen

→ Implement a formal AI procurement checklist: vendors must provide conformity documentation, security attestations, and bias testing evidence

→ Annual AI portfolio review: assess risk tier reassignments, system retirements, and emerging risks from model updates or scope creep

→ Establish whistleblower protection for employees reporting AI-related harms or policy violations in NB public sector bodies

→ **SME handover obligation:** When the engagement ends, provide the SME client with a written AI System Summary — risk tier, data processed, what the AI does, what to monitor, who to call if something goes wrong

*§ 5 · APPLIED USE CASE GUIDANCE*

# NB-Specific AI Use Case Guidance

### 5A · SENTIMENT ANALYSIS IN NB CONTEXTS

*RISK TIER*

Typically Tier 2 (Limited Risk) for citizen feedback analysis. Escalates to Tier 1 (High Risk) if outputs influence eligibility decisions, law enforcement prioritization, or employment decisions.

— Validate accuracy on NB-specific French and English corpora — Acadian French dialect patterns differ significantly from Parisian French training data

— Test for sarcasm, colloquialisms, and code-switching patterns (Chiac) common in bilingual NB communities

— Establish minimum F1 score thresholds per language and demographic segment before deployment — document parity gaps

— Do not use sentiment scores as sole input to any consequential decision about a person

— Inform data subjects when their communications are sentiment-scored (PIPEDA / NB Privacy Act obligations)

— Establish a human review pathway for outputs scored at confidence thresholds below 0.75

— **Prohibited:** No real-time sentiment analysis of NB residents in public spaces without explicit consent and legal authority

— **Prohibited:** No sentiment scoring of children's communications in education contexts without specific ethical review

### 5B · AI CHATBOTS IN NB PUBLIC AND GOVERNMENT SERVICES

*RISK TIER*

Typically Tier 2 (Limited Risk) for information and service navigation chatbots. Escalates to Tier 1 if the chatbot influences eligibility, benefits entitlement, or clinical referral decisions.

— Display bilingual AI disclosure at chatbot launch — not buried in footer terms

— Offer clear escalation path to human agent at any point in the conversation

— Disclose data retention practices: how long chat logs are stored, by whom, for what purpose

— For RAG-based chatbots: validate retrieval sources against authoritative GNB content; implement document freshness controls

— Implement prompt injection defenses — especially critical for public-facing chatbots processing untrusted user input

— Rate-limit and monitor for abuse patterns (data harvesting, jailbreak attempts)

— Do not store personally identifiable information in chatbot session logs beyond operational necessity

*§ 5C · BIAS CLASSIFICATION & MITIGATION*

# Understanding and Addressing AI Bias in NB Deployments

BiasinAI systemsmanifests atmultiplestages ofthemachine learning pipeline. Thissection drawson **Amii's Principled AI** framework and **IBM's AI Fairness 360 (AIF360)** to give NB deployers a structured approach to bias identification and mitigation.

### THE SIX PIPELINE BIAS TYPES (AMII / SURESH & GUTTAG)

| BIAS TYPE | WHERE IT ENTERS | NB EXAMPLE | MITIGATION |
|---|---|---|---|
| Historical | Training data reflecting past societal inequities | Hiring AI trained on historical NB decisions may perpetuate underrepresentation of Acadian, First Nations, or rural candidates | Audit training data provenance; apply reweighting or resampling |
| Representation | Training data underrepresents certain populations | Sentiment analysis trained on English social media data underperforms on Acadian French text | Supplement with NB French-language corpora; set minimum language-parity thresholds |
| Measurement | Proxy variables less accurate for some groups | Using postal code as creditworthiness proxy encodes rural NB economic disadvantage | Validate proxy variables for differential accuracy; document all proxy choices |
| Aggregation | Single model applied to heterogeneous populations | Healthcare triage AI applied uniformly without accounting for distinct First Nations health profiles | Stratify models or use group-specific calibration; sub-group performance analysis |
| Evaluation | Benchmarks don't represent deployment population | Chatbot evaluated only on standard English queries fails on Chiac (NB code-switching) inputs | Build evaluation datasets representative of NB user language patterns |
| Deployment | Model used in a context it was not designed for | Sentiment tool validated for customer service applied to social services caseworker interactions | Scope AI systems narrowly; flag any scope expansion as a new deployment requiring fresh assessment |

### IBM AI FAIRNESS 360 (AIF360) — BIAS MITIGATION PIPELINE

| STAGE | WHEN APPLIED | AIF360 TOOLS | NB MINIMUM REQUIREMENT |
|---|---|---|---|
| Pre-processing | Before model training — on the dataset | Optimized Preprocessing, Reweighing, Disparate Impact Remover | Required for Tier 1; recommended for Tier 2 NLP and sentiment systems |
| In-processing | During model training | Adversarial Debiasing, Meta-Fair Classifier, Learning Fair Representations | Required for Tier 1 systems where training pipeline is within the deployer's control |
| Post-processing | After prediction — on model outputs | Equalized Odds Post-Processing, Reject Option Classification, Calibrated Equalized Odds | Required for ALL Tier 1 and Tier 2 systems — this stage is always within deployer control |

> ***NB-SPECIFIC BIAS PRIORITY: LANGUAGE REPRESENTATION***
>
> Given NB's unique status as Canada's only officially bilingual province, representation bias and evaluation bias related to French and English language parity are the highest-priority bias risks for any NLP, sentiment analysis, or conversational AI system deployed in NB. No such system should be approved for Tier 1 or Tier 2 deployment without documented evidence of equivalent accuracy across both official languages — including validation against Acadian French dialect patterns, not just standard Parisian French training corpora.

*§ 5D · CRITICAL INFRASTRUCTURE*

# AI in New Brunswick's Critical Infrastructure

New Brunswick isnot justaparticipantinCanada's criticalinfrastructureecosystem — it is increasingly a national hub for it. AI in critical infrastructure represents the highest-stakes deployment context in this framework.

> *LIVE LEGISLATIVE CONTEXT — BILL C-8 / CCSPA (2026)*
>
> Bill C-8 passed the House of Commons on March 26, 2026, enacting the Critical Cyber Systems Protection Act. CI operators in NB — including energy, telecommunications, transportation, and finance — now face mandatory cyber security program requirements and incident reporting obligations. AI systems in these sectors must meet both this NB framework and Bill C-8's federal obligations.

### NB'S CRITICAL INFRASTRUCTURE AI ECOSYSTEM

| INSTITUTION / ASSET | RELEVANCE TO AI IN NB CI |
|---|---|
| Siemens Cybersecurity Centre for Critical Infrastructure | Fredericton — global OT/ICS security research and deployment |
| Thales National Digital Excellence Centre (NDEC) | Fredericton — critical infrastructure cyber defence |
| Canadian Institute for Cybersecurity (CIC) at UNB | Canada's first CI cybersecurity research institute — 50+ researchers |
| NBCC CI-SOC | Saint John — Canada's first CI Security Operations Centre for IT/OT convergence training |
| NB Power Smart Grid | AI-enabled grid modernization with Siemens; NERC CIP standards enforced by NB EUB |

### AI RISK SUB-CLASSIFICATION FOR CI CONTEXTS

| SUB-TIER | USE CASE | SECTOR EXAMPLES | KEY OBLIGATIONS |
|---|---|---|---|
| 1-A · Critical | Autonomous control of physical systems | Grid switching, water treatment dosing, pipeline pressure management | Human-in-loop mandatory for ALL consequential actions; formal safety case; OT penetration testing; IEC 62443 alignment |
| 1-B · High | AI-driven threat detection | SOC AI, SIEM ML, network behavioural analytics | Model drift monitoring; false positive/negative SLAs; human analyst review before automated response; quarterly red-team testing |

| SUB-TIER | USE CASE | SECTOR EXAMPLES | KEY OBLIGATIONS |
|---|---|---|---|
| 1-B · High | Predictive maintenance AI | Failure prediction, maintenance scheduling for energy/transport/water | Explainability required; human sign-off before acting on AI-generated deferrals |
| 2 · Limited | Administrative AI in CI organizations | Procurement AI, HR screening, financial forecasting | Standard Tier 2 obligations plus elevated supply chain scrutiny |

### FIVE CI-SPECIFIC AI SECURITY REQUIREMENTS

01 **OT Architecture Segregation.** AI inference must not occur on the same network segment as live control systems without a validated air-gap or unidirectional gateway. Apply IEC 62443 as the baseline.

02 **AI Supply Chain Risk Management.** Maintain a complete AI Bill of Materials (AI-BOM). No AI component from an unvetted vendor may be deployed without a formal security assessment. AI supply chain incidents are reportable under Bill C-8.

03 **Adversarial Attack Preparedness.** Test AI systems against prompt injection, model inversion, membership inference, data poisoning, evasion attacks, and model extraction. Red-team exercises: annually for Tier 1-B, semi-annually for Tier 1-A.

04 **AI Failure Mode & Continuity Planning.** Every AI system in CI must have a documented and tested manual override procedure. Operational plans cannot assume AI availability during a cyber incident.

05 **AI Incident Reporting.** AI-related CI incidents must be reported to CCCS via My Cyber Portal within Bill C-8 timeframes. Also notify CyberSecure NB and, for energy, the NB EUB. Post-incident reviews within 30 days.

*§ 6 · GOVERNANCE STRUCTURE*

# NB AI Governance Architecture

| BODY | MANDATE |
| --- | --- |
| NB AI Policy & Standards Office (Proposed — GNB) | Develop and maintain the NB AI Framework; publish mandatory AI procurement standards; maintain a public NB AI Systems Registry for Tier 1 and Tier 2 systems; issue guidance on emerging AI risks including generative AI and agentic systems |
| Office of the Information & Privacy Commissioner of NB (Expanded Mandate) | Review AI-related DPIAs for Tier 1 systems; receive and investigate complaints from NB residents regarding AI decisions; coordinate with OPC on cross-jurisdictional deployments. Note: binding AI-specific enforcement powers require legislative expansion of the IPC's mandate — this framework recommends that expansion. |
| CyberSecure NB (Technical Authority) | Publish and maintain AI-specific cybersecurity technical standards; provide threat intelligence on emerging AI attack vectors; conduct or certify security assessments of Tier 1 AI systems; support incident response when AI systems are involved in cybersecurity events |

## CONSULTANT PRACTICE STANDARDS

> *WRITTEN BY A PRACTITIONER, FOR PRACTITIONERS*
>
> This framework was drafted by an NB GRC and AI Trust practitioner who works in the same consultancy ecosystem it describes. The following are not punitive obligations — they are the professional baseline that any responsible AI consultant in NB should already be applying, offered as a shared standard for what clients should expect from any AI engagement in NB.

— Complete a risk tier classification for every AI system proposed or deployed — documented and discussed with the client before work begins

— Provide clients with a plain-language AI System Summary covering: purpose, risk tier, data inputs, key limitations, and escalation contacts — in both official languages where the client serves bilingual populations

— Evaluate proposed AI vendors against NB framework requirements and document findings in the SOW or contract schedule

— Notify the client promptly upon discovery of any AI security incident, model failure, or evidence of bias that affects NB residents or their data

— Do not recommend, deploy, or white-label AI systems that cannot provide training data provenance, security documentation, or bias testing results — if a vendor cannot answer these questions, that is itself a risk finding

— Where a client engagement involves First Nations communities or data, apply OCAP® principles and recommend Nation-to-Nation consultation before any AI system is scoped, not after it is built

— **SME handover obligation:** When the engagement ends, provide the SME client with a written AI System Summary they can actually use after you leave

*§ 7 · IMPLEMENTATION ROADMAP*

# Phased Adoption Timeline

Arealistic,sequenced implementationplan. The roadmap reflects the current federal policy environment — no binding Canadian AI statute is in force as of mid-2026 — and is calibrated toward voluntary adoption and stakeholder consultation as the near-term levers.

Summer-Fall 2026 Current Phase

### Open Consultation & Community Feedback

Publish framework publicly. Invite feedbackfrom NB tech sector, GNB staff, privacy advocates, First Nations governance bodies, labour organizations, and civil society. Initiate Nation-to-Nation outreach to Wolastoqiyik and Mi'kmaq nations. Seek academic peer review from UNB, Université de Moncton, and Mount Allison.

Fall 2026 Policy Submission

### Formal Submission — Revised Draft v2.0

Submit revisedframework incorporating consultation feedback to GNB, NBIF, and federal AI strategy consultation channels as a model for sub-national AI governance in bilingual Canadian jurisdictions. Publish final bilingual version in English and French.

Q1 2027 Voluntary Adoption

### Toolkits & Pilot Programs

Releaseself-assessmenttoolkit, risk classification guide, and Model Card template. Pilot with 3–5 GNB departments. Engage NBCC and UNB for AI ethics curriculum alignment.

Q2–Q3 2027 Mandatory Standards

### GNB Procurement Standards

AIprocurement checklistbecomesacondition of GNB technology RFPs. CyberSecure NB publishes AI security baseline. IPC begins accepting AI-related DPIA filings. First Nations consultation findings incorporated into v2.0.

Q4 2027 Public Registry

### NB AI Systems Registry Launch

PublicregistryofTier1and Tier2AIsystems deployed by GNB goes live. Annual transparency report published. Incident reporting mechanism activated.

2028 Biennial Review

### First Framework Review — Scope Expansion

ReviewagainstfederalAIstrategy implementation and EU AI Act enforcement lessons. Expand scope to cover environmental harms, labour displacement, and youth AI safety — areas explicitly deferred from v1.1.

§ 8 · REFERENCES & FRAMEWORK ALIGNMENT

# Framework Alignment Map

| SOURCE | KEY ELEMENTS REFERENCED |
|---|---|
| EU AI Act (2024/1689) — International | Risk tier architecture (Articles 5, 6, 50) — adapted, not adopted wholesale; Transparency (Article 13); Human oversight mandate (Article 14) |
| NIST AI RMF 1.0 — USA / NIST | GOVERN, MAP, MEASURE, MANAGE functions; AI risk identification; Trustworthiness characteristics taxonomy. Explicit pillar mapping in §8b below. |
| ISO/IEC 42001:2023 — International Standard | AI management system requirements; Organizational accountability controls; AI impact assessment methodology |
| Federal AI Strategy (2026) — Canadian Federal | AIDA / Bill C-27 died on order paper January 2025 — successor expected; ISED National AI Sprint Report (February 2026); Evan Solomon appointed Canada's first Minister for AI and Digital Innovation (May 2025) |
| Bill C-8 / CCSPA (2026) — Canadian Federal | Critical Cyber Systems Protection Act — passed House of Commons March 26, 2026; Mandatory cyber security programs for designated CI operators; CCCS Critical Infrastructure Resilience initiative (April 2026) |
| Directive on Automated Decision-Making — Federal Treasury Board | Algorithmic impact assessment tool; Decision automation levels 1–4; Notice, explanation, and appeal rights; Peer review and audit requirements |
| OCAP® Principles & UNDRIP — Indigenous Rights & Data | OCAP® — First Nations Information Governance Centre (Ownership, Control, Access, Possession); UNDRIP Act (Canada, 2021); FPIC |
| India SUVAS — Bilingual AI Governance Reference | Human-in-the-loop post-editing model for AI translation; domain-specific training corpus approach; continuous feedback loop for accuracy improvement — governance architecture applicable to NB bilingual AI requirements |
| New Brunswick Legislation — Provincial | Protection of Personal Information Act; NB Human Rights Act; Official Languages Act; NB Energy and Utilities Board Act; CyberSecure NB / CBDC Program |
| Amii Principled AI & IBM AIF360 — Bias & Fairness | Suresh & Guttag six pipeline bias types; IBM AI Fairness 360 — 70+ fairness metrics and 10 mitigation algorithms; Amii Datasheets for Datasets, Model Cards, and Algorithmic Impact Assessments |
| SANS AI Security Maturity Model™ (May 2026) | Three pillars: Protect AI, Utilize AI, Govern AI. Five maturity stages. Agentic AI, Non-Human Identity, and the Principle of Least Agency. Explicit NB stage mapping in §8c below. |

### § 8B · NIST AI RMF 1.0 — EXPLICIT PILLAR MAPPING

The NIST AI Risk Management Framework organizes AI risk management into four core functions: **GOVERN, MAP, MEASURE,** and **MANAGE**. The following maps each of this framework's five pillars to the NIST AI RMF function it primarily addresses.

| NB PILLAR | PRIMARY NIST FUNCTION | SECONDARY | KEY NIST ALIGNMENT |
|---|---|---|---|
| Pillar 1 — Cybersecurity-First AI Design | MANAGE | MAP | MANAGE 1.0 (risks prioritized and addressed); MAP 1.0 (context established, risks identified); threat modelling and penetration testing align to MANAGE 2.0 risk treatment actions |
| Pillar 2 — Honest AI Interaction Standards | GOVERN | MANAGE | GOVERN 1.0 (policies for trustworthy AI); GOVERN 4.0 (commitment to transparency); disclosure requirements align to NIST Trustworthiness characteristic: Transparency |
| Pillar 3 — Fairness & Bias Management | MEASURE | MAP | MEASURE 2.0 (AI risks analyzed, including bias and fairness); MAP 5.0 (impacts on individuals, communities); bias incident register aligns to MEASURE 4.0 (risk measurement results documented) |
| Pillar 4 — Explainability & Auditability | MEASURE | GOVERN | MEASURE 1.0 (AI risk measurement approaches identified); GOVERN 6.0 (policies for transparency and accountability); audit logs and XAI align to NIST Trustworthiness: Explainability and Accountability |
| Pillar 5 — Governance & Human Oversight | GOVERN | MANAGE | GOVERN 1.0–6.0 (full governance function); AI Accountability Owner maps to GOVERN 2.0; human-in-the-loop requirements align to MANAGE 3.0 |

***NIST AI RMF FUNCTION DEFINITIONS***

GOVERN: Cultivates organizational practices that make AI risk management a priority — policies, culture, roles, accountability structures.

MAP: Categorizes AI risks in context — AI system identification, stakeholder engagement, risk identification across the AI lifecycle.

MEASURE: Analyzes and assesses AI risks quantitatively and qualitatively — bias evaluation, performance monitoring, risk measurement documentation.

MANAGE: Prioritizes and addresses AI risks — risk treatment, incident response, and residual risk monitoring.

### § 8C · SANS AI SECURITY MATURITY MODEL™ — NB ALIGNMENT

The **SANS AI Security Maturity Model™** (released May 12, 2026) provides a five-stage, three-pillar operational roadmap from ad hoc AI use to fully governed and secured AI programs. Where the SANS model tells NB organizations how mature their security posture is, this framework tells them what governance obligations apply at each tier.

### The Three SANS Pillars — Mapped to NB Framework

| SANS PILLAR | SANS DEFINITION | NB FRAMEWORK ALIGNMENT |
|---|---|---|
| Protect AI | Securing AI systems from external and internal threats — model integrity, data pipeline security, inference endpoint protection, adversarial attack resilience | Pillar 1 (Cybersecurity-First AI Design); §5d CI requirements 1–4; threat modelling, penetration testing, OT architecture segregation, AI-BOM |
| Utilize AI | Governing how AI tools are adopted — shadow AI, approved model registries, data governance for AI inputs, responsible use policies | Pillar 2 (Transparent Disclosure); SME Recipient Obligations (§1); foundation model governance requirement; data residency requirements |
| Govern AI | Organizational accountability, risk management, compliance, and oversight — policies, risk registers, audit trails, incident response, regulatory reporting | Pillar 5 (Governance & Human Oversight); Pillar 4 (Explainability & Auditability); AI Accountability Owner; AI Review Board; Bill C-8 incident reporting |

### The Five SANS Maturity Stages — NB Context and Obligations

| STAGE | SANS DEFINITION | NB ORGANIZATION PROFILE | MINIMUM NB OBLIGATIONS |
|---|---|---|---|
| Stage 1 Unaware / Ad Hoc | No formal AI security policy. AI tools adopted without visibility or governance. Sensitive data moving into public models without controls. | Most NB SMEs receiving CyberSecure NB grants. Organizations deploying first AI chatbot or sentiment tool without a governance framework. | Complete SME Baseline Obligations (§1). Conduct risk tier classification immediately. Apply Tier 3 minimum obligations. Do not proceed to Tier 1 or Tier 2 deployments without uplift. |
| Stage 2 Aware / Reactive | AI use is visible but governance is reactive. Some policies exist but not enforced consistently. | NB mid-sized organizations with an IT function but no dedicated AI governance role. GNB departments with informal AI use but no AI procurement standard. | Implement Pillar 1 and Pillar 2. Assign an AI Accountability Owner. Establish a Bias Incident Register. Apply Tier 2 obligations to all chatbot and sentiment deployments. |

| STAGE | SANS DEFINITION | NB ORGANIZATION PROFILE | MINIMUM NB OBLIGATIONS |
|---|---|---|---|
| Stage 3 Defined / Managed | Formal AI security program in place. Risk assessments conducted. Controls documented and measured. Red teaming at least quarterly. | NB critical infrastructure operators (energy, health, finance). GNB departments with formal digital governance. Organizations under Bill C-8 obligations. | All five pillars active. Full Tier 1 conformity assessment complete. AI-BOM maintained. Bias audits documented. Audit logs retained per Pillar 4. Incident reporting operational. |
| Stage 4 Optimized | Continuous AI risk monitoring. Adversarial simulation in production. Regulatory evidence generated automatically. Agent-to-agent interactions logged and audited. | NB CI operators targeting full Bill C-8 compliance and EU AI Act audit readiness. Organizations with agentic AI deployments. | Apply Principle of Least Agency to all agentic AI systems: each agent has a documented identity, scoped permissions per task, and session-level audit trail. NHI governance required for all AI agents. |
| Stage 5 Optimizing / Adaptive | AI security posture updates in near real-time. Security controls tested continuously via adversarial simulation. Organization contributes to industry standards. | Aspirational for NB CI hub organizations (Siemens, Thales, CIC/UNB). NB organizations providing AI security services to other sectors. | All NB framework obligations met. Contribute findings to NB AI Systems Registry and proposed NB AI Policy Office. Share threat intelligence with CyberSecure NB and CCCS. |

***AGENTIC AI — THE NB GOVERNANCE GAP TO WATCH***

The SANS AI Security Maturity Model identifies agentic AI and Non-Human Identity (NHI) as the area where most organizations have a real blind spot. The Principle of Least Agency — the agentic counterpart to least privilege — requires that every AI agent in production be granted only the minimum permissions required for the specific task, with a documented identity, scoped authorization, and a session-level audit trail. NB organizations deploying agentic AI systems (AI that takes autonomous actions, calls external APIs, or orchestrates other AI systems) must apply this principle as a non-negotiable security control. This will be incorporated as a formal requirement in v2.0.